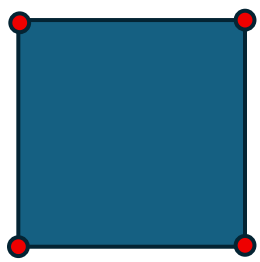
SHORT DESCRIPTION:

This website allows users to be able to visualize how 2D geometric transformations work in real time, by allowing users to manipulate a square by translating and scaling it in real time through user input.

HOW IT WORKS:

The square is made up of four points/corners,



Represented by four matrices:

[corner1_x corner1_y] [corner2_x corner2_y]

[corner3_x corner3_y] [corner4_x corner4_y]

It is then connected made into vectors (differentiated via colors), and filled in a semi transparent blue.

The transformation is done by applying this formula to each corner of the square:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Where $\begin{bmatrix} x \\ y \end{bmatrix}$ represents the original corner,

$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$ represents how much the user wants to scale the square,

and finally, $\begin{bmatrix} t_x \\ t_y \end{bmatrix}$ represents where the user wants to place the square,

$\begin{bmatrix} x' \\ y' \end{bmatrix}$ represents the new x and y coordinates of the corner.

ROTATION:

Rotation spins the square around its center point by a given angle. The rotation is done via the standard 2D rotation matrix formula:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Where θ represents the angle the user wants the square to be rotated at,

and $\begin{bmatrix} x \\ y \end{bmatrix}$ represents each corner's position relative to the center of the square.

Because the rotation formula always rotates around the origin point (0, 0), we need to follow a 3 important rules:

1.) **Translate to origin:** Subtract the center from every corner of the square, so that the center of the square is now (0, 0).
2.) **Apply the rotation matrix:** Multiply the shifted corners by the rotation matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

to spin the corners around (0, 0)
3.) **Putting it back:** Add the center back to every corner of the square so that it returns to its original position, now being rotated in place.

IMPORTANT THINGS TO NOTE:

Note that with translating the square, as x increases, it moves to the right. And as x

decreases, it moves to the left.
And as y increases, it moves to the the bottom. And as y decreases, it moves to the top.

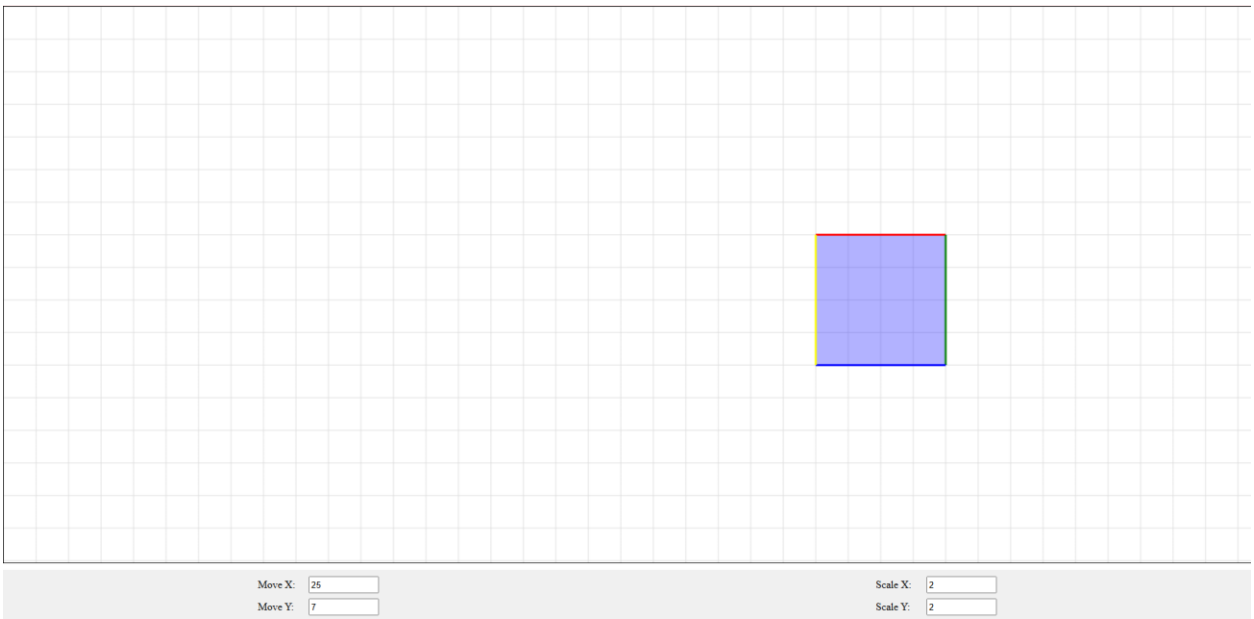


Figure 1- Square at x: 25, y: 7

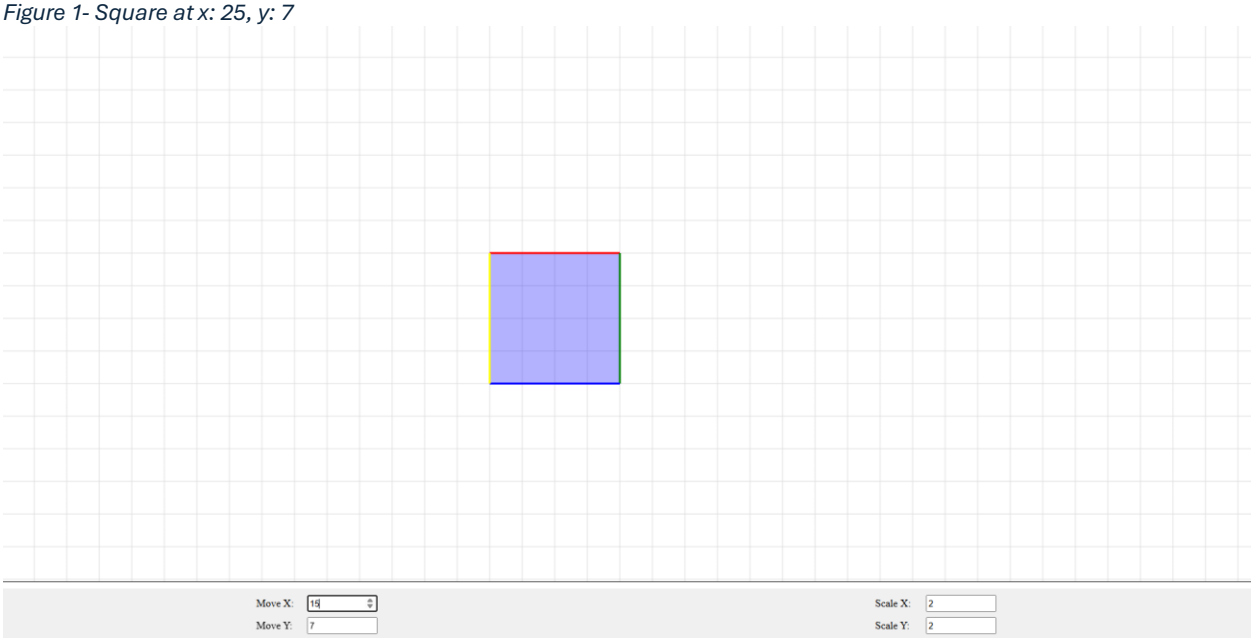


Figure 2 - Square at x: 15, y: 7

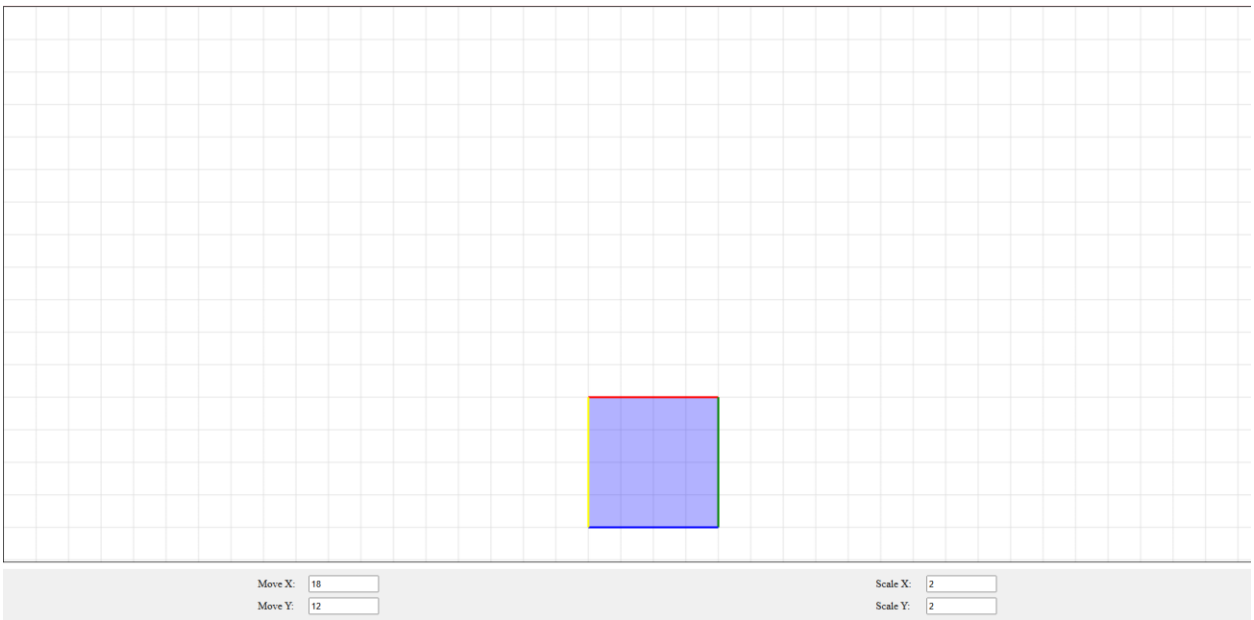


Figure 3 - Square at x: 18, y: 12

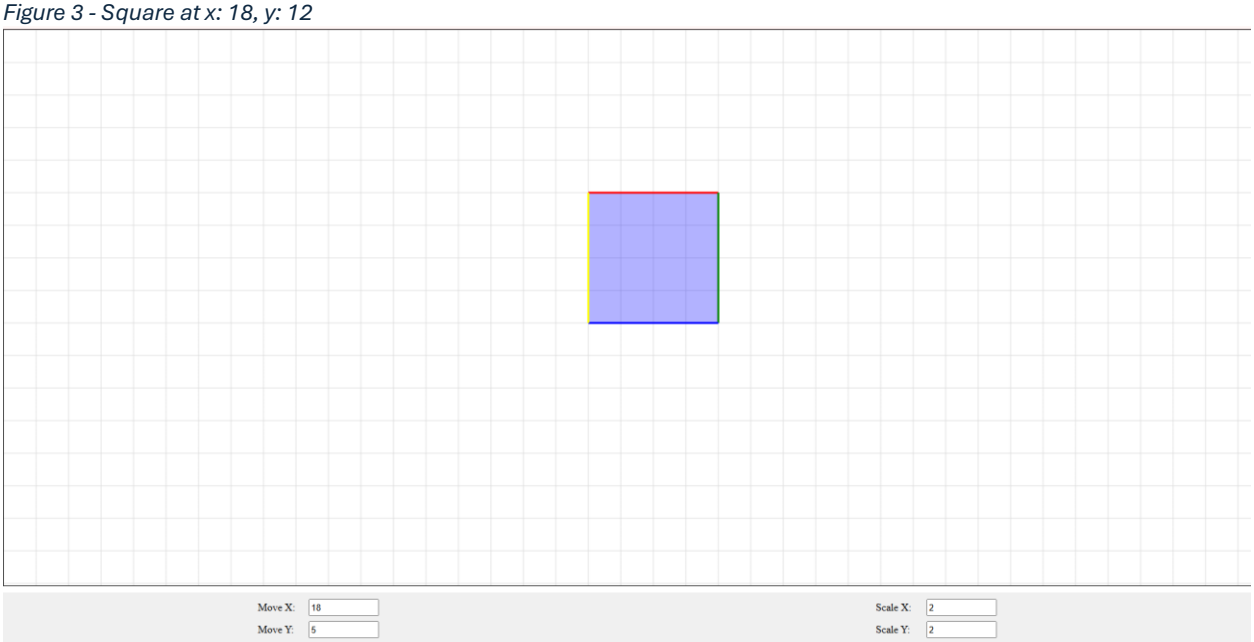


Figure 4 - Square at x: 18, y: 5

Additionally, for scaling values, as x increases, it grows to the right. However, if the value of x is negative, the sqaure is flipped and grows to the left.

And as y increases, it grows to the bottom, and if the value of y is negative, the square is flipped and grows to the top.

This happens because scaling multiplies the coordinates. Positive values stretch the shape in the same direction, while negative values reverse the direction of the coordinates, causing the shape to flip and grow in the opposite direction.